ECE 329 Exam 3 · FRONT: waves $\mu_0 = 4\pi\times10^{-7}$ | $\epsilon_0 = \frac{10^{-9}}{36\pi}$ | $\eta_0 = 120\pi \approx 377$ | $c = 0.3$ m/ns | $\mu_0 c = \eta_0$, $\epsilon_0 c\eta_0 = 1$

Phasors, no calculator

$X(t) = \mathrm{Re}\{\tilde{X}e^{j\omega t}\}$ | $\partial_t \to j\omega$ | $\cos \to 1$, $\sin \to -j$, $-\cos \to -1$

$\frac{1}{j} = -j$ | $\sqrt{j} = e^{j45^\circ} = \frac{1+j}{\sqrt{2}}$ | $\sqrt{-j} = \frac{1-j}{\sqrt{2}}$ | $e^{j\pi} = -1$

$\cos x = \frac{e^{jx}+e^{-jx}}{2}$ | $\sin x = \frac{e^{jx}-e^{-jx}}{2j}$

divide: × conjugate | polar: sizes ÷, angles −

$\langle \mathrm{Re}\{Ae^{j\omega t}\}\, \mathrm{Re}\{Be^{j\omega t}\}\rangle = \frac{1}{2}\mathrm{Re}\{AB^*\}$

triangles 1:1:$\sqrt{2}$ → 45° | 1:$\sqrt{3}$:2 → 30°, 60° | 3:4:5 → 36.9°, 53.1°

travel: $e^{-j\beta z}$ or $\omega t - \beta z \to +z$

Lossy medium

$x = \frac{\sigma}{\omega\epsilon} = \frac{60\sigma\lambda_0}{\epsilon_r}$ (conduction ÷ displacement): $x < 0.1$ low loss, $x > 10$ conductor, else exact

method: $k = \omega\sqrt{\mu\epsilon}$, $\eta_i = \sqrt{\mu/\epsilon}$; $1 - jx = \rho e^{-j\theta}$

$\gamma = jk\sqrt{\rho}\, e^{-j\theta/2}$ | $\eta = \frac{\eta_i}{\sqrt{\rho}} e^{j\theta/2}$ | $\tau = \frac{\theta}{2} \le 45^\circ$

check: $\angle\gamma + \angle\eta = 90^\circ$, $|\gamma||\eta| = \omega\mu$

exact: $\beta^2 - \alpha^2 = k^2$ | $2\alpha\beta = \omega\mu\sigma$

conductor: $\delta = \frac{1}{\alpha} = \frac{1}{\sqrt{\pi f \mu\sigma}}$ | $\gamma = \frac{1+j}{\delta}$ | $\eta = \frac{1+j}{\sigma\delta}$

inside: $\lambda = 2\pi\delta$, $v_p = \omega\delta$; at $z = \delta$: field 37%, power 13.5%

H lags E by τ; both decay as $e^{-\alpha z}$

Reading a wave

$E_0 e^{-\alpha z}\cos(\omega t - \beta z + \phi)$: $\lambda = \frac{2\pi}{\beta}$, $v_p = \frac{\omega}{\beta}$, $\frac{1}{\alpha}$ = the $\frac{1}{e}$ distance

H from E: size ÷ $|\eta|$, phase $-\tau$, direction $\hat{k} \times \hat{e}$

$\tan\tau = \frac{\alpha}{\beta}$ (because $\angle\gamma = 90^\circ - \tau$)

Cu $\sigma = 5.8\times10^7$: $\delta \approx 66\ \mu$m at 1 MHz, $\delta \propto \frac{1}{\sqrt{f}}$ | sea: $\sigma \approx 4$, $\epsilon_r \approx 81$

Power and loss

$\langle S\rangle = \frac{E_0^2}{2|\eta|}\cos\tau\, e^{-2\alpha z} = \frac{1}{2}|H_0|^2\,\mathrm{Re}\{\eta\}\, e^{-2\alpha z}$

lossless $\frac{E_0^2}{2\eta}$ | field $e^{-\alpha z}$, power $e^{-2\alpha z}$

dB = $10\log\frac{P_1}{P_2} = 8.69\,\alpha z$ | 3 dB = ½, 10 dB = $\frac{1}{10}$, 20 dB = $\frac{1}{100}$

heat per volume $\frac{1}{2}\sigma|\tilde{E}|^2 = -\nabla\cdot\langle\mathbf{S}\rangle$

$R_s = \frac{1}{\sigma\delta} = \sqrt{\frac{\pi f\mu}{\sigma}}$ | conductor $\eta = (1+j)R_s$

loss/area $\frac{1}{2}R_s|\tilde{J}_s|^2$, $\tilde{\mathbf{J}}_s = \hat{n}\times\tilde{\mathbf{H}}$ (mirror: $|\tilde{J}_s| = 2|\tilde{H}_i|$); mirror absorbs $\frac{4R_s}{\eta_0}$

pressure: mirror $\frac{2\langle S_i\rangle}{c}$, absorber $\frac{\langle S_i\rangle}{c}$ (N/m²)

$\langle w_e\rangle = \frac{1}{4}\epsilon|\tilde{E}|^2$, $\langle w_m\rangle = \frac{1}{4}\mu|\tilde{H}|^2$; lossless: equal, $\langle S\rangle = v_p(\langle w_e\rangle + \langle w_m\rangle)$

$-\nabla\cdot\mathbf{S} = \partial_t(w_e + w_m) + \mathbf{J}\cdot\mathbf{E}$: both terms; on average only $\mathbf{J}\cdot\mathbf{E}$ is left

Polarization

linear: in phase or 180° | circular: equal sizes, ±90° | else elliptical

hand: $\tilde{\mathbf{E}} = \mathbf{a} + j\mathbf{b}$ at one plane: $\mathbf{E}$ turns $\mathbf{a} \to -\mathbf{b}$; spin axis $\mathbf{b}\times\mathbf{a}$ along travel: RH, against: LH

right thumb along travel, fingers turn with $\mathbf{E}$ → RH

$+z$: $\hat{x} - j\hat{y}$ RH | $\hat{x} + j\hat{y}$ LH | $\cos\psi\,\hat{x} + \sin\psi\,\hat{y}$ RH ($\psi = \omega t - \beta z$)

reflection (real Γ) flips the hand

$\langle S\rangle = \frac{|\tilde{E}_s|^2 + |\tilde{E}_b|^2}{2\eta}$: the two parts add

Current sheets

sheet at z_s: $\tilde{\mathbf{E}} = -\frac{\eta}{2}\tilde{\mathbf{J}}_s e^{-jk|z-z_s|}$ both sides (lossy $e^{-\gamma|z-z_s|}$): E even

$\tilde{\mathbf{H}} = \frac{1}{2}\tilde{\mathbf{J}}_s \times \hat{n}\, e^{-jk|z-z_s|}$, $\hat{n}$ sheet → point: H odd, jumps J_s

want $\tilde{\mathbf{E}}_0$ beside it: $\tilde{\mathbf{J}}_s = -\frac{2}{\eta}\tilde{\mathbf{E}}_0$ | per side

$\langle S\rangle = \frac{\eta}{8}|\tilde{J}_s|^2$

two sheets: add, each with its own $|z - z_k|$; between them $e^{-jkz} + e^{+jkz} \to \cos$, $\sin$

equal $\tilde{\mathbf{J}}_s$ at 0 and d, between:

$\tilde{\mathbf{E}} = -\eta\tilde{\mathbf{J}}_s e^{-jkd/2}\cos k(z - \frac{d}{2})$,

$\tilde{\mathbf{H}} = -j(\tilde{\mathbf{J}}_s \times \hat{z})e^{-jkd/2}\sin k(z - \frac{d}{2})$; beyond d: $-\frac{\eta}{2}\tilde{\mathbf{J}}_s(1 + e^{jkd})e^{-jkz}$

gap $\frac{\lambda}{2}$, same current: 0 outside | gap $\frac{\lambda}{4}$, $\tilde{J}_2 = -j\tilde{J}_1$: all toward sheet 2's side

Reflection, standing waves

$\tilde{E}_r = \Gamma\tilde{E}_i$, $\tilde{H}_r$ gets $-\Gamma$ | $\eta_2 < \eta_1 \Rightarrow \Gamma < 0$

power: back $|\Gamma|^2$, through $1 - |\Gamma|^2 = \frac{\eta_1}{\eta_2}|\tau|^2$

$|\tilde{E}_1| = E_0|1 + \Gamma e^{-j2\beta\ell}|$ (ℓ from boundary): max $(1 + |\Gamma|)E_0$, min $(1 - |\Gamma|)E_0$, $\frac{\lambda}{4}$ apart; $\Gamma < 0$: min at the boundary

PEC at $z = 0$: $\mathbf{E}_1 = 2E_0\sin\beta z\, \sin\omega t\,\hat{x}$ |

$\mathbf{H}_1 = \frac{2E_0}{\eta}\cos\beta z\cos\omega t\,\hat{y}$ | $\mathbf{J}_s = \frac{2E_0}{\eta}\cos\omega t\,\hat{x}$ | $\langle S\rangle = 0$

E nodes every $\frac{\lambda}{2}$ from the wall; H max at the wall

$\frac{\lambda}{4}$ layer with $\eta_2 = \sqrt{\eta_1\eta_3}$: no reflection | $\frac{\lambda}{2}$ slab: invisible

Line from the fields

$\mathcal{L} = \frac{\mu}{GF}$ | $\mathcal{C} = \epsilon GF$ | $G = \sigma GF$ | $\mathcal{LC} = \mu\epsilon$ | $\frac{G}{C} = \frac{\sigma}{\epsilon}$

GF: plates $\frac{W}{d}$ | coax $\frac{2\pi}{\ln(b/a)}$ | twin lead $\frac{\pi}{\cosh^{-1}(D/2a)}$

coax $Z_0 = \frac{\eta}{2\pi}\ln\frac{b}{a} = \frac{60}{\sqrt{\epsilon_r}}\ln\frac{b}{a}$ | plates $Z_0 = \eta\frac{d}{W}$

plates $V = E_x d$, $I = H_y W$ | coax $V = \int_a^b E_r\,dr$, $I = 2\pi r H_\phi$

$Z_0 = \frac{1}{v\mathcal{C}} = v\mathcal{L}$ | $\mathcal{C} = \frac{Q'}{V}$, $\mathcal{L} = \frac{\Psi'}{I}$ (per meter)

Launch, reflect, steady state

$V^+ = \tau_g V_g$ (line looks like Z_0) | $I^\pm = \pm\frac{V^\pm}{Z_0}$ | current reflects with $-\Gamma$

Γ: matched 0, short −1, open +1; $R > Z_0 \Rightarrow \Gamma > 0$

$V_L = (1+\Gamma_L)V^+$ | $I_L = (1-\Gamma_L)\frac{V^+}{Z_0}$

joint $Z_a \to Z_b$: $\Gamma = \frac{Z_b - Z_a}{Z_b + Z_a}$, $\tau = 1+\Gamma$; from b's side: $-\Gamma$, $1-\Gamma$

two lines after the joint: $Z_b \| Z_c$ | resistor at the joint: $R \| Z_b$

steady state (DC): the line vanishes: $V = V_g\frac{R_L}{R_g+R_L}$, $I = \frac{V_g}{R_g+R_L}$; $= \tau_g V_g\frac{1+\Gamma_L}{1-\Gamma_L\Gamma_g}$

C load: short first (−1), open at the end (+1) | L: open first, short at the end

Bounce diagrams

t down, z across, slopes $\pm v$; write each wave's value on its line

$V(z,t)$ = sum of every wave whose front has passed z by t

arrivals at z: $\frac{z}{v}$, $\frac{2l-z}{v}$, $\frac{2l+z}{v}$, $\frac{4l-z}{v}$, ...

snapshot at t: fronts at $z = vt$, $2l - vt$, ...; first in time = farthest in z

pulse of width w: $u(t) - u(t-w)$, length vw on the line

impulse: spikes $\tau_g(\Gamma_L\Gamma_g)^n$ (the sheet's sums)

scope at $z = 0$: level $\tau_g V_g$; jump at $2T$ is $\tau_g V_g \Gamma_L(1+\Gamma_g) \to \Gamma_L$; $T \to l = vT$

junctions: each its own Γ, τ; every transmitted wave starts new bounces

Phasor lines

$Z_{in} = Z_0\frac{Z_L + jZ_0\tan\beta l}{Z_0 + jZ_L\tan\beta l}$

$\frac{\lambda}{4}$: $Z_{in} = \frac{Z_0^2}{Z_L}$ (inverts) | $\frac{\lambda}{2}$: $Z_{in} = Z_L$

match R_L to R_{in}: $\frac{\lambda}{4}$ of $Z_0 = \sqrt{R_{in}R_L}$ (at that f, and odd multiples)

$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0}$; $\Gamma(d) = \Gamma_L e^{-j2\beta d}$ turns clockwise toward the generator

$|\tilde{V}|$ max where $\Gamma(d)$ is real and > 0: $2\beta d_{max} = \theta_L + 2\pi n$; min $\frac{\lambda}{4}$ away

real $R_L > Z_0$: max at load, VSWR $= \frac{R_L}{Z_0}$ | $R_L < Z_0$: min at load, VSWR $= \frac{Z_0}{R_L}$

$P_L = \frac{|\tilde{V}^+|^2}{2Z_0}(1 - |\Gamma_L|^2)$ = power in (lossless)

$\tilde{V}^+ = \frac{\tilde{V}(l)}{e^{j\beta l}(1+\Gamma(l))}$

Stubs and resonance

short: $Z = jZ_0\tan\beta l$: under $\frac{\lambda}{4}$ → L; 0 at $l = n\frac{\lambda}{2}$, ∞ at $l = (2n-1)\frac{\lambda}{4}$

open: $Z = -jZ_0\cot\beta l$: under $\frac{\lambda}{4}$ → C; ∞ at $n\frac{\lambda}{2}$, 0 at $(2n-1)\frac{\lambda}{4}$

in f ($\beta l = \frac{2\pi f l}{v}$): short stub 0 at $f = \frac{nv}{2l}$, ∞ at $f = \frac{(2n-1)v}{4l}$; open: swap

short, $\beta l \ll 1$: $L = \mathcal{L}l$ | open: $C = \mathcal{C}l$

shunt stub that is 0 at f: notch (kills f) | ∞ at f: invisible | series stub ∞: blocks

ringing line: same ends (S-S, O-O) $f = \frac{nv}{2l}$, period $\frac{2l}{v}$ | mixed (S-O) $f = \frac{(2n-1)v}{4l}$, period $\frac{4l}{v}$

Speed rules and numbers

1. lossless numbers first, then $x = \frac{\sigma}{\omega\epsilon}$, then the matching sheet line

2. $\tilde{\mathbf{H}} = \frac{1}{\eta}\hat{k} \times \tilde{\mathbf{E}}$ | $\tilde{\mathbf{E}} = \eta\tilde{\mathbf{H}} \times \hat{k}$ ($\hat{k}$ = travel)

3. sign of Γ first: smaller η or R ahead ⇒ $\Gamma < 0$ ⇒ minimum at the boundary or load

4. $\frac{\lambda}{4}$ inverts, $\frac{\lambda}{2}$ repeats; short under $\frac{\lambda}{4}$ = L, open under $\frac{\lambda}{4}$ = C

5. last bounce level = the DC divider: check it

6. a joint: V continuous and currents add (that is $\tau = 1+\Gamma$)

7. three τ's: angle of η | $1+\Gamma$ | $\tau_g = \frac{Z_0}{R_g+Z_0}$: name which one

$e^{-1} = .368$ | $e^{-2} = .135$ | $e^{-3} = .050$ | $e^{-4} = .018$ | $e^{-5} = .0067$

$\ln 2 = .693$ | $\ln 10 = 2.30$ | $\sqrt{2} = 1.414$ | $\sqrt{3} = 1.732$ | $\pi^2 \approx 9.87$

300 MHz: $\lambda_0 = 1$ m, $k_0 = 2\pi$ | $\omega\mu_0 = k_0\eta_0$ | $\omega\epsilon_0 = \frac{k_0}{\eta_0}$

units: α Np/m, β rad/m, η Ω, S W/m², J_s A/m, $\mathcal{L}$ H/m, $\mathcal{C}$ F/m; 1 Np = 8.69 dB